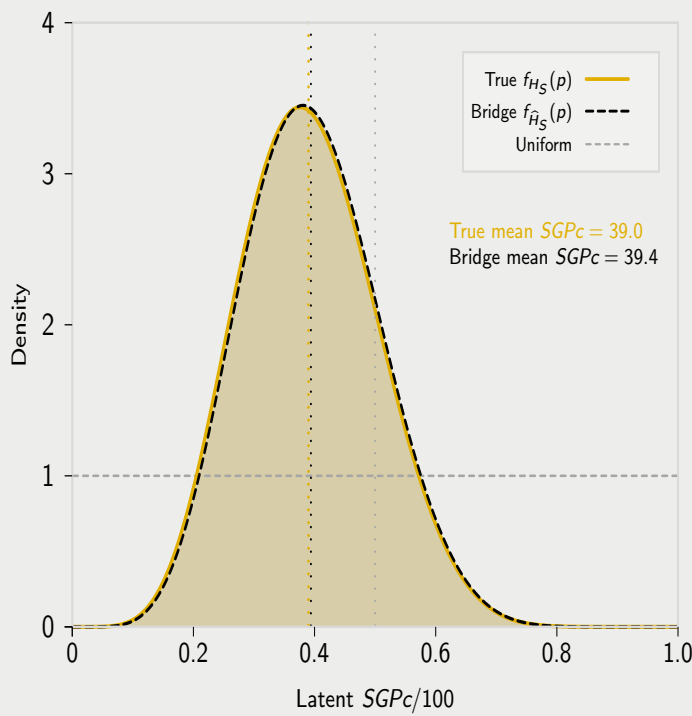


Inference Error Budget for Copula-Induced Stochastic Growth Regimes

Accuracy and Precision of Mean *SGPc* Estimated Without Linked Data

Synthetic validation: True mean *SGPc* = 39.0, bridge estimate = 39.4, MAE = 0.42 across 25 conditions, baseline *t*-copula ($\rho = 0.72$, $df = 8$)

A. Define the linked-data benchmark
True subgroup mean *SGPc* from longitudinal pairs



Suppose longitudinal linkage is available: each student i in subgroup S has a prior quantile U_i and a current quantile V_i . With these linked pairs, we can compute the true latent conditional percentiles

$$P_i = F_0(V_i | U_i),$$

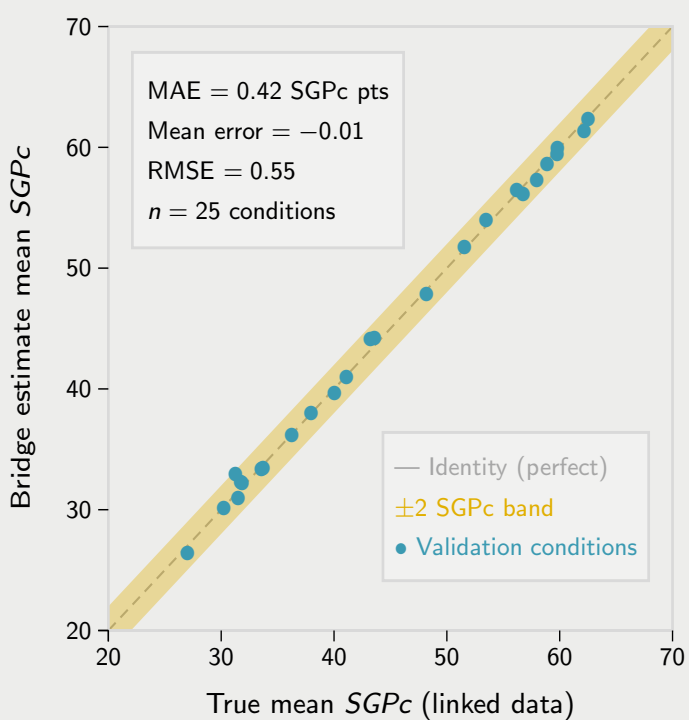
where F_0 is the conditional CDF from the baseline copula C_0 . The *truth benchmark* for the subgroup is

$$\theta_S^{\text{true}} = 100 \mathbb{E}(P_S).$$

This is the quantity we would report if individual-level linkage existed. Panel A shows the density $f_{H_S}(p)$ of the true growth regime alongside the bridge-recovered density $f_{\hat{H}_S}(p)$. The vertical lines mark their respective means. The fundamental role of Panel A is to *define what accuracy means*: accuracy can only be assessed relative to a well-defined truth. Without a linked-data benchmark, “accuracy” becomes internal consistency rather than genuine error measurement.¹

The small displacement between the amber (true) and black (bridge) densities is the *bridge cost*: the price paid for erasing individual-level linkage and relying on the copula bridge instead.

B. Forget linkage and re-infer
Bridge accuracy: unlinked estimate versus truth



Panel B operationalizes *bridge accuracy* across 25 validation conditions. For each condition r :

- Compute the truth θ_S^{true} from linked data.
- Erase the pairings; retain only the unlinked marginals F_U and F_V .
- Re-run the copula bridge (STEP 3) to obtain θ_S^{bridge} .
- Record the bridge error:

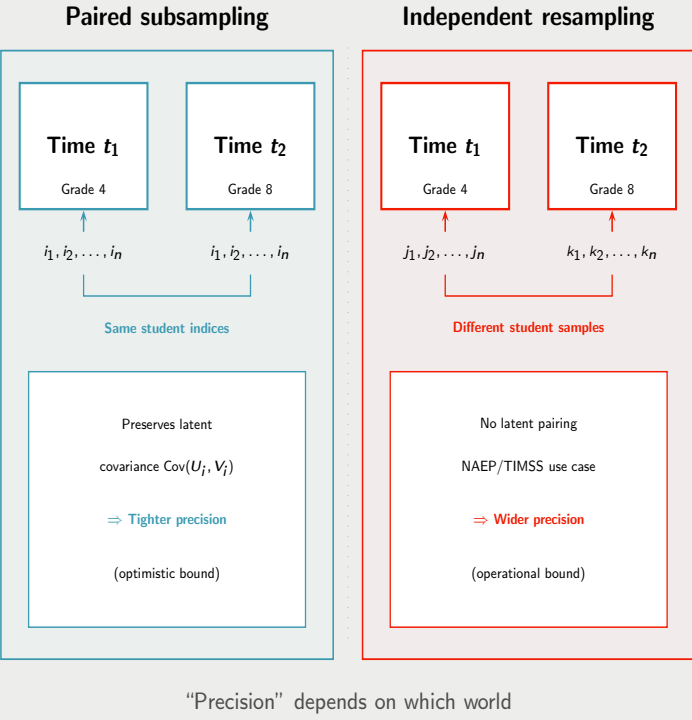
$$\text{Bridge error}_r = \theta_S^{\text{bridge}} - \theta_S^{\text{true}}.$$

The scatter plot shows *truth vs. bridge estimate* for each condition. Points near the identity line indicate high accuracy. The amber band marks ± 2 SGPc points from perfect agreement.

Accuracy is summarized by center displacement, not by interval width. The relevant metrics are mean absolute error (MAE = 0.42), mean signed error (−0.01), and RMSE (0.55), all on the 0–100 SGPc scale.

This validation answers: *how far does the unlinked bridge move us from the truth that linked data would provide?*²

C. Repeat the sampling design
Paired subsampling vs. independent resampling



“Precision” is not one thing. It depends on what is being resampled.

Paired subsampling. Sample the *same* student indices at both time points: draw (U_i, V_i) pairs by resampling indices $\{i_1, \dots, i_n\}$ from the subgroup. This preserves the latent covariance $\text{Cov}(U_i, V_i)$ and typically yields *tighter* repeated-sampling precision.

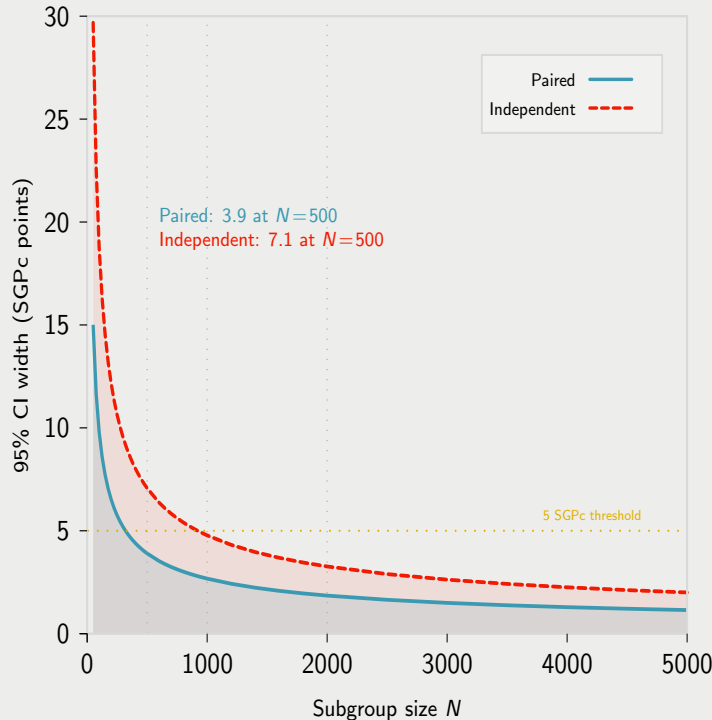
Independent cross-sectional resampling. Sample grade- t_1 and grade- t_2 populations independently: different students contribute to F_U and F_V . This matches the NAEP/TIMSS use case and produces *wider* uncertainty because no within-student covariance anchors the bridge.

Without distinguishing these two worlds, readers will conflate:

- linked-subsample stability (optimistic),
- unlinked-cross-sectional operating precision (realistic), and
- bridge accuracy (a separate axis).

Those are three different beasts. Panel C makes the design dependence visible so that the precision curves in Panel D can be read correctly.³

D. Quantify precision by N
Operating uncertainty as a function of subgroup size



Precision quantifies the repeated-sampling spread of the bridge estimator $\hat{\theta}_S$ as a function of subgroup size N . Panel D shows the 95% confidence interval width under both resampling designs from Panel C:

$$\text{Precision} = \text{CI width}_{95}(\hat{\theta}_S) = 3.92 \cdot \text{SE}(\hat{\theta}_S).$$

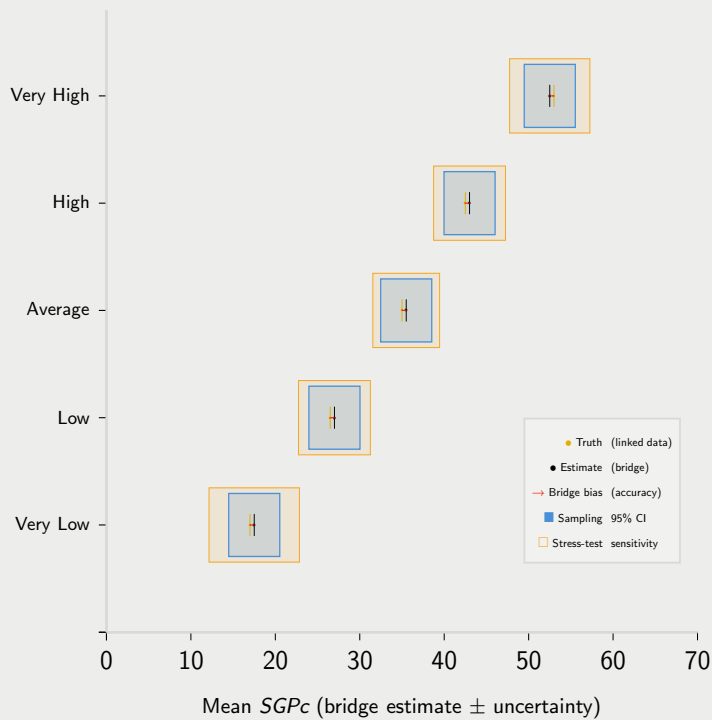
Two curves appear:

- Paired** (solid): the optimistic bound from bootstrap subsampling with linked indices. At $N = 500$: CI width ≈ 3.9 SGPc points.
- Independent** (dashed): the operational bound matching NAEP/TIMSS-style cross-sectional designs. At $N = 500$: CI width ≈ 7.1 SGPc points.

Precision improves sharply from small N into the mid-range (200–800), then flattens. The amber dotted line at 5 SGPc points marks a practical threshold: below this width, subgroup mean *SGPc* becomes operationally useful for reporting and comparison.

The gap between the two curves is the *design effect*: the additional uncertainty from not having linked cohorts. This gap is largest at small N and shrinks (absolutely) as N grows, but the *ratio* remains roughly constant.⁴

E. Report total uncertainty
Combined error, interval, and category stability



The final panel answers the operational question: *what total uncertainty attaches to a reported subgroup mean SGPc, and how stable are resulting classifications?*

For each of five representative subgroups (at $N = 500$, independent design), Panel E displays nested uncertainty layers:

- Truth** (amber dot): θ_S^{true} from linked data.
- Bridge estimate** (black dot): θ_S^{bridge} . The **arrow** between truth and estimate is the bridge bias (accuracy cost).
- Sampling band**: 95% CI from repeated sampling (± 3.5 SGPc).
- Stress-test band**: additional sensitivity to copula family choice, $P \perp U$ relaxation, and classification perturbation.

The total uncertainty envelope combines:

$$\underbrace{\text{bridge bias}}_{\text{Panel B}} + \underbrace{\text{sampling precision}}_{\text{Panel D}} + \underbrace{\text{model sensitivity}}_{\text{stress tests}}.$$

Action thresholds and category labels are meaningful only within this combined envelope. A subgroup classified as “low growth” at $\hat{\theta}_S = 35$ is robust if the full band stays below 40; it is fragile if the band spans 30–45.⁵

¹ In the synthetic validation setting, the true growth regime is $P_S \sim \text{Beta}(7.0, 11.0)$ with mean *SGPc* = 39.0 and $\kappa = 18$, under a baseline *t*-copula ($\rho = 0.72$, $df = 8$). The bridge-recovered estimate uses the same copula family and Beta regime class, evaluated by Wasserstein-1 minimum distance.

² Bridge accuracy here refers to the displacement introduced solely by erasing individual-level linkage and relying on marginal-only information to estimate the growth regime. It does *not* include sampling variability (that is precision, Panel D) or model-choice sensitivity (stress tests, Panel E).

³ The paired-vs-independent distinction matters operationally because most large-scale assessment programs (NAEP, TIMSS, PIRLS) use independently drawn cross-sectional samples. Precision estimates based on paired subsampling of a longitudinal cohort are systematically tighter than what users of the method will experience in the field.

⁴ The precision curves assume the baseline copula and regime family are correctly specified. Model misspecification introduces additional uncertainty not captured by these curves; see the stress-test component in Panel E. At large N (> 2,000), bridge accuracy (Panel B) typically dominates sampling precision as the primary error source.

⁵ Category stability is assessed by asking: would the subgroup's classification change under perturbation of the estimate by one standard error, by the bridge bias, or by the stress-test envelope? Robust classifications survive all three perturbations. Fragile classifications require either larger samples, tighter bridge accuracy, or wider reporting bands.

⁶ **Model-choice stress tests** (right margin of Panel E band): copula family sensitivity (Frank vs. *t*), relaxation of the $P \perp U$ assumption, and classification changes under perturbation. These are second-order uncertainty components that bound the “unknown unknowns” of the copula bridge.